

Pore-Space-Partition-Oriented Sandwich Platinum Array Confined in a Metal–Organic Framework for Boosting Overall Water Splitting

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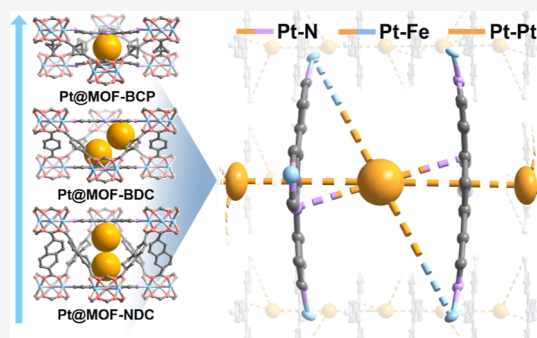
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ABSTRACT: The atomic-level dispersion of noble metals in porous materials is prospective for highly efficient catalyst systems but is still inscrutable. Inspired by sandwich compounds, platinum (Pt) atoms were rationally confined within a metal–organic framework (MOF) through pore space partition by conjugate interactions between Pt and parallel-distributed aromatic pore partitioners, forming 1D infinite Pt arrays. By regulating the distance between adjacent aromatic rings from 7 to 4.5 to 3 Å ($\sim$ the diameter of the Pt atom), the particle size in sandwich Pt arrays decreased in a linear fashion with the Pt electronic orbital simultaneously regulated by a pore partitioner, MOF nodes, and near noble atoms, as revealed by AC high-angle annular dark-field scanning transmission electron microscopy, X-ray absorption spectroscopy, and theoretical simulations. The optimized MOF

pore endows the Pt@MOF-BCP material with highly efficient utilization of Pt sites, best electron transmission efficiency, and excellent catalytic stability. At 10 mA·cm^{−2}, an optimal Pt@MOF-BCP catalyst only needs ultralow overpotentials of 2.5 mV for hydrogen evolution reaction in the acidic electrolyte and 265 mV for oxygen evolution reaction in the alkaline electrolyte, exhibiting exceptional difunctional electrocatalytic performance superior to most benchmark catalysts. Remarkably, the cell voltage of Pt@MOF-BCP was dramatically reduced to 1.47 V at 10 mA·cm^{−2} in practical application.



INTRODUCTION

Water splitting is considered as an environment-friendly energy conversion for mitigating the global energy and environmental crisis.¹ Hydrogen evolution reaction (HER) and oxygen evolution reaction (OER) are the cores of electrocatalytic water splitting.² However, the sluggish kinetics of two reactions limit energy conversion efficiency.³ To realize the high catalytic efficiency during water electrolysis, efficiency catalysts that can minimize overpotential are desirable.⁴ To date, noble metals are still considered to be the most efficient commercial electrocatalyst for water splitting.⁵ However, high cost, low durability, and scarcities of noble metal catalysts plagued their large-scale application in water splitting.⁶

Construction of noble metal-based catalysts with precise atomic-level control over size, structure, and composition is the most effective strategy to balance cost and catalytic efficiency.⁷ At the same time, the interactions of noble metals and support materials can promote effective dissociation of reactants and optimize the adsorption/desorption of key intermediates.⁸ Several strategies have been developed to anchor noble metals including defect engineering, interface inducing, and pore encapsulating. Nevertheless, the adjustable size, loading amount, stability, and defined atomic structure still remain formidable challenges.⁹

Organic linker-protected noble metals with customized metal size, transparent structure, and explicit interaction force are typical for molecular catalysts.¹⁰ Especially, sandwich organometal compounds with a metal atom lying between two aromatic organic rings can display exceptional electrochemical stability and catalytic performance.¹¹ Parallel aromatic rings with a π hole feature are utilized as electronic acceptors for crystal engineering,¹² and the center metal atoms serve as donor typically,¹³ forming the metals- π -conjugated system. Benefiting from the dominant interactions with high bond energy, center noble metals are always stabilized extremely.¹⁴ Moreover, the d-electronic orbitals of the metal overlap with the π electronic orbit of organic rings in sandwich structures and may effectively regulate the electronic structure of catalysts and enhance the electron transport rate.¹⁵

A metal–organic framework (MOF) possessing vast types of aromatic linkers and porous structures is regarded as the ideal model support to encapsulate noble metals at the atomic

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level.¹⁶ Notably, the dimension of encapsulated noble metals is mainly regulated by the MOF pore; however, the feeble confined operation usually results in a low loading amount, torpid electron transport efficiency, and poor catalytic efficiency. The atomic-level dispersion of noble metals in MOF pores has been a huge challenge up to now.

Pore space partition (PSP), an efficient strategy, has been proposed to maximize the utilization efficiency of MOF channels/pores. PSP in *acs*-type metal–organic frameworks (*pac*s-MOFs) through the induction of triangular aromatic partitioners into one-dimensional (1D) hexagonal channels formed by ditopic linkers effectively promoted their gas adsorption and separation ability.¹⁷ The original continuous 1D hexagonal channels are precisely partitioned into small finite segments by tuning the pore-partitioning agents and ditopic linkers.¹⁸ Building upon previous advances, the vertical pore dimension of each finite segment can be finely modulated by adjusting the length of the ditopic linkers.¹⁹ Herein, the *pac*s-MOF platform is demonstrated as an ideal platform to precisely load Pt metals. The aromatic partitioners with a π hole lying in a 1D hexagonal MOF channel are parallel to each other with a regulatable separation distance, which rationally anchor Pt atoms and isolate them in the confined pores. The sizes of trapped Pt clusters are engineered by regulating ditopic linkers, and aromatic pore partitioners act as a π electron donor with Pt metals to generate unique 1D infinite Pt arrays with a clear structure and composition. Benefiting from the tandem $\pi\cdots\text{Pt}\cdots\pi$ electronic conjugate system, Pt@MOF catalysts exhibited enhanced catalytic activity and stability. Specially, Pt atoms confined in a MOF-BCP skeleton with a comparable pore size of 3 Å ($\sim$ the diameter of the Pt atom) can interact with not only aromatic partitioned linkers but also MOF metal nodes, which strongly promoted the atomic utilization and electron transmission efficiency of Pt. The optimal Pt@MOF-BCP catalyst at 10 mA·cm^{−2} only needs overpotentials of 2.5 mV for HER in the acidic electrolyte and 265 mV for the OER in alkaline electrolyte together with a cell voltage as low as 1.47 V for practical overall water splitting.

RESULTS AND DISCUSSION

A series of FeNi-MIL-88 bimetallic MOF precursors²⁰ with outstanding synergistic catalytic effects were first designed to implement the precise Pt encapsulation (Figure 1). The continuous 1D hexagonal channels were then partitioned into small finite segments precisely along the *c* axis via a 2,4,6-tri(4-pyridinyl)-1-pyridine (TPP) pore partitioner. Sandwich π -electron spaces between two parallel TPP linkers were then formed to stabilize Pt atoms. To further control the size of trapped Pt atoms, different ditopic linkers of varying lengths, including bicyclo[1.1.1]pentane, 1,4-terephthalic acid, and naphthalene-2,6-dicarboxylic acid (denoted as BCP, BDC, and 2,6-NDC) are adopted. The corresponding MOF precursors, designated as MOF-BCP, MOF-BDC, and MOF-NDC, with the similar structural configuration as Cambridge Crystallographic Data Centre (CCDC) deposition numbers of 2159092,²¹ 2043323,²² and 2043329,²² respectively, exhibit the sandwich π -electron space heights of about 3.0 Å, 4.5 Å, and 7 Å (the heights correspond to the distances between two adjacent TPP planes with the van der Waals radii taken into account), respectively (Figure 1).²² Notably, the pore sizes are just 1-fold, 1.5-fold, and 2-fold the diameter of the single Pt atom ($\sim$ 3 Å), promoting the atomic-level dispersion-like

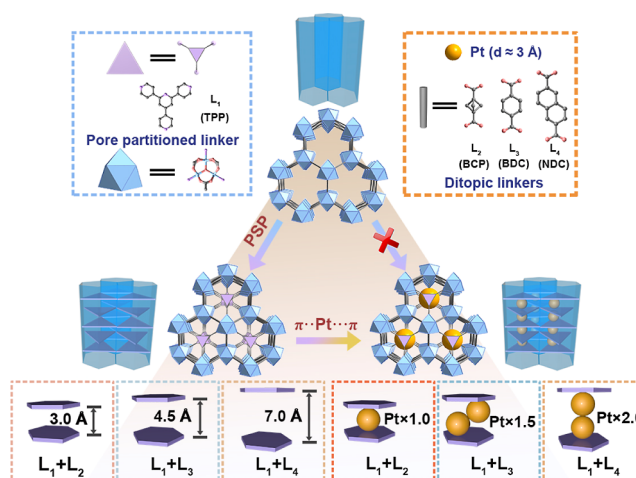


Figure 1. Schematic illustration of sandwich Pt arrays rationally confined in a MOF oriented by PSP.

sandwich organometal compounds. Each segment space contains two parallel TPP, six metal trinuclear clusters, and six ditopic linkers. The high density of exposed nitrogen sites provided by the TPP linkers effectively confines the Pt within each segment. Overall, the aromatic pore partitioner alignments and suitable pore space in these MOFs facilitate the atomic-level Pt encapsulation and lead to the formation of a series of catalysts denoted as Pt@MOF-BCP, Pt@MOF-BDC, and Pt@MOF-NDC (Figure 1).

Density functional theory (DFT) calculations further confirmed the interactions and electron density distributions of these Pt@MOF catalysts (Figures 2 and S1).²³ The Pt atoms confined within the MOF pores were used in DFT-optimized models to approximate the behavior of Pt. As expected, confined in limited partitioned space, the electronic orbits of the Pt atom overlapped the π electronic orbit of aromatic TPP linkers,²⁴ giving $\pi\cdots\text{Pt}\cdots\pi$ electronic conjugated coordination. Optimal Pt atoms can locate in the tandem pore segments between parallel TPP partitioners, forming an infinite $\cdots\text{Pt}\cdots\text{TPP}\cdots\text{Pt}\cdots\text{TPP}\cdots$ sandwich array structure. Derived from the stronger electronegativity and higher electron density of the pyridine N atom, favorable Pt–N coordination bonds further stabilized the Pt array arranged between Pt atoms and TPP linkers (Figures 2a and S2–S4).²⁵

Charge redistribution between Pt and the MOFs could be clearly observed (Figure 2b). Due to the electronegativity difference of Pt (2.28) < C (2.55) < N (3.04),²⁶ electron accumulation around N and C and electron depletion around Pt were demonstrated obviously. The movement of an electron from metal Pt to a nonmetallic TPP linker indicated that the TPP organic linker served as an electronic acceptor and Pt was regarded as a donor. Notably, due to the best encapsulation space, the electronic orbit of Pt and MOF-BCP overlapped the furthest, exhibiting an obvious charge redistribution phenomenon, indicating the strong interaction between Pt and MOF frameworks.²⁷ A two-dimensional electron localization function (ELF) of Pt@MOF-BCP exhibited a stronger interaction and shorter interatomic distance between Pt–N, further manifesting the strongly electron overlap in Pt@MOF-BCP (Figure 2c).²⁸

The theoretical lengths of Pt–N bonds in Pt@MOF-BCP/BDC/NDC were 1.94, 2.04, and 2.04 Å with the adsorption energies between the Pt atom and MOFs of −2.56 eV, −1.31

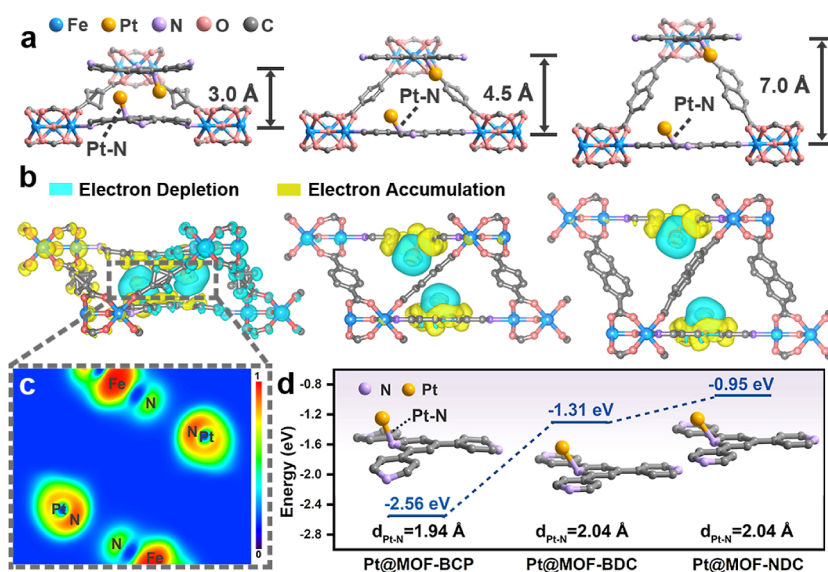


Figure 2. (a) Computed position of Pt atoms in MOF structures. (b) Electron density distribution of Pt@MOFs. Yellow and cyan represent the accumulation and depletion of electron density, respectively. Atomic color schemes include Fe (blue), O (red), C (gray), N (purple), and Pt (orange). (c) ELF of Pt@MOF-BCP. (d) Bond length of Pt–N in Pt@MOFs and corresponding adsorption energy.

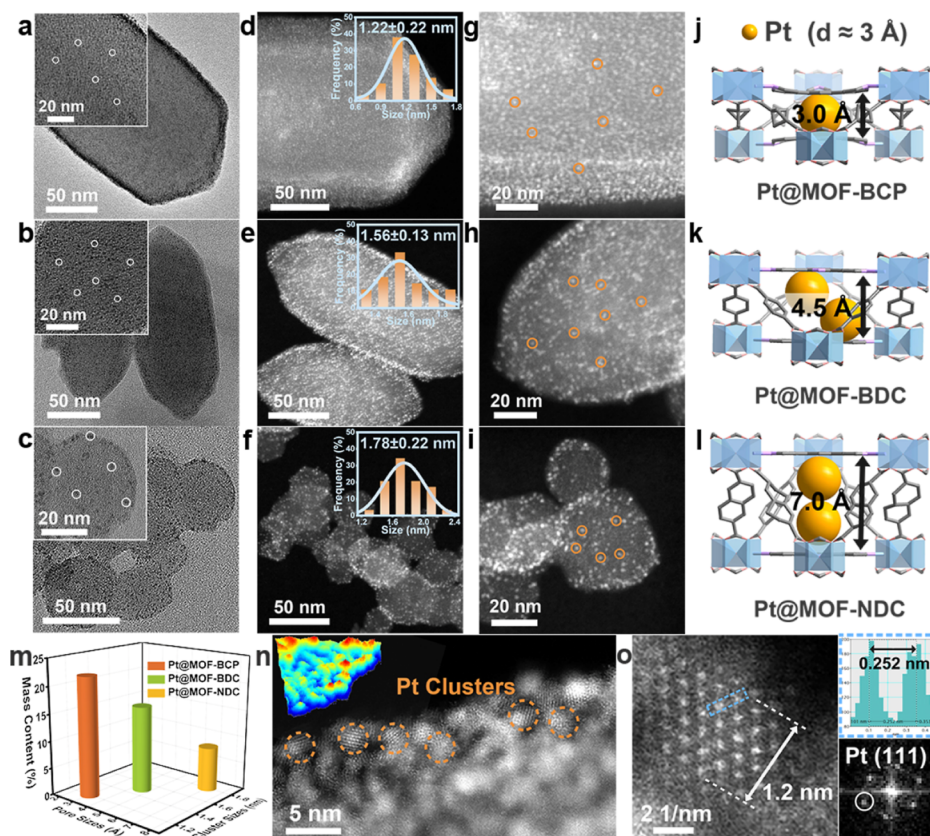


Figure 3. (a–c) TEM images, (d–f) HAADF-STEM images, (g–i) magnified HAADF-STEM images, and (j–l) corresponding structural models of Pt@MOFs. (m) Quantify Pt clusters of the series Pt@MOFs. (n–o) AC HAADF-STEM images and FFT image of Pt@MOF-BCP.

eV, and -0.95 eV, respectively (Figure 2d). All these results indicated that Pt was successfully encapsulated in the pore of MOFs by PSP and interacted with parallel pore partitioners, forming $\pi \cdots \text{Pt} \cdots \pi$ electronic conjugated coordination.²⁷ Among these, Pt@MOF-BCP with the optimal pore size possessed the strongest host–guest interaction and most stable

$\pi \cdots \text{Pt} \cdots \pi$ electronic conjugated coordination and thus can optimize the electronic structure extremely.

Powder X-ray diffraction (PXRD) patterns of MOF precursors were all in good agreement with the simulated patterns from single crystallographic data (Figures S5–S7). The main diffraction peaks of MOF-BCP, MOF-BDC, and MOF-NDC appeared at 9.48° , 8.33° , and 7.39° , respectively,

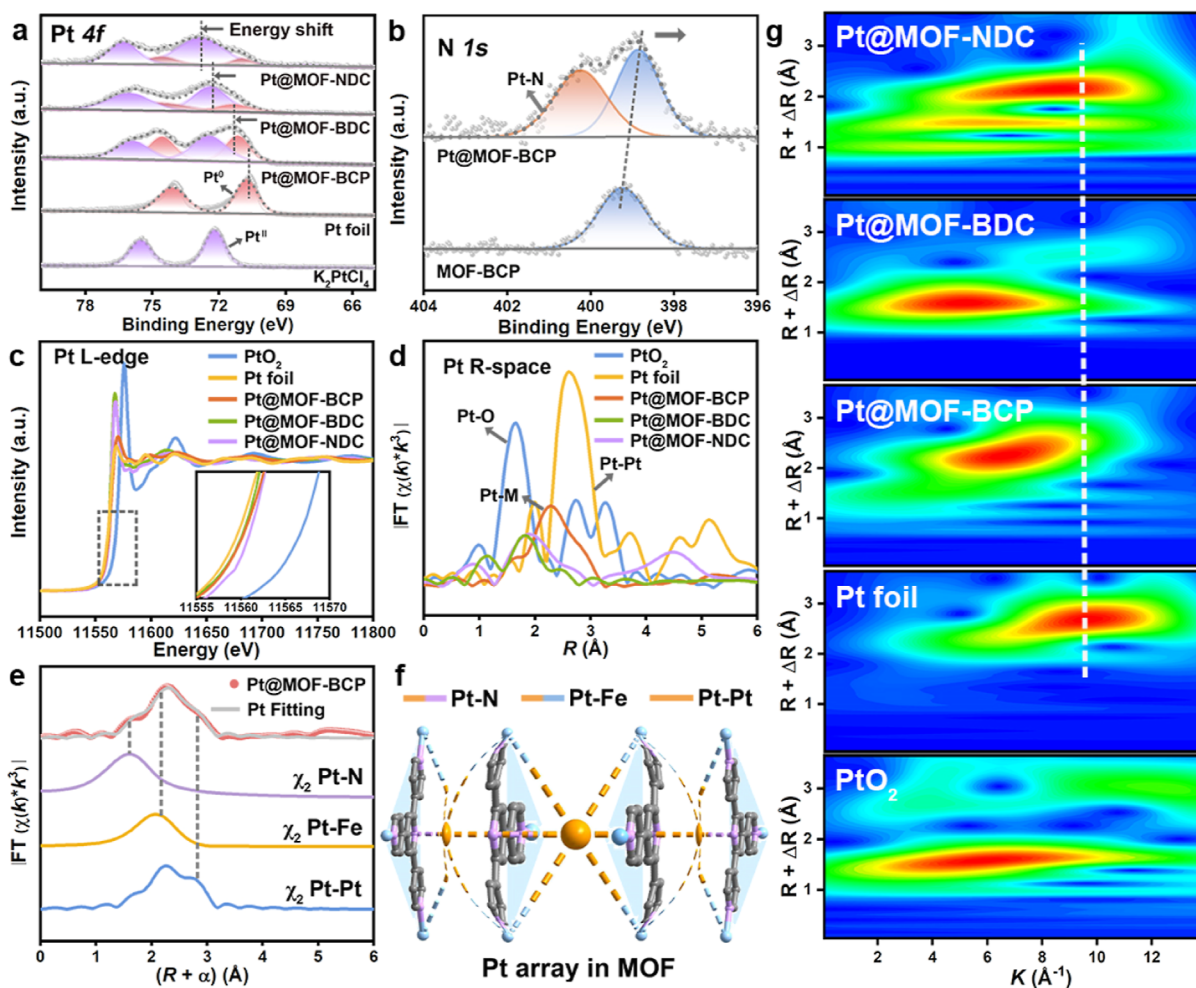


Figure 4. (a) XPS spectra for Pt@MOFs, K_2PtCl_4 , and Pt foil of Pt 4f. (b) XPS spectra for Pt@MOF-BCP and MOF-BCP of N 1s. (c) Normalized Pt L-edge XANES spectra. (d) Corresponding FT-EXAFS spectra. (e) EXAFS fitting and quantitative least-squares EXAFS curve-fitting. (f) Schematic of sandwich Pt array in Pt@MOFs. (g) Wavelet transforms for the k_3 -weighted EXAFS.

all corresponding to the (101) planes of their previously reported structures.^{21,22} Following the subsequent introduction and reduction reaction of Pt, the PXRD patterns of the Pt@MOFs remained well, suggesting that the MOF frameworks remained intact and their crystallinity was well retained (Figures S8–S10). On the other hand, MIL-88 MOF without pore space partitioners exhibited lower operation stability and failed to anchor Pt atoms into the 1D channels (Figures S11–S14). Scanning electron microscopy (SEM) images of the pore-space-partitioned MOF-BCP, MOF-BDC, and MOF-NDC precursors exhibited their spherical and spindle-like morphology with 50 ~ 200 nm (Figures S15–S17). SEM and transmission electron microscopy (TEM) images clearly indicated that Pt@MOF catalysts well maintain their regular morphologies (Figures 3a–c and S18–S20).

High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of Pt@MOFs revealed abundant dispersive bright spots, representing the heavier Pt clusters (Figure 3d–i). The magnified HAADF-STEM images displayed that Pt atoms were successfully packaged into MOF pores uniformly without agglomeration. Due to the interconnected pore space, Pt clusters were formed. The Pt cluster sizes were linearly regulated by the smallest axis of the pore spaces with average sizes of 1.22 ± 0.22 , 1.56 ± 0.13 , and 1.78 ± 0.22 nm for Pt@MOF-BCP/BDC/NDC (Figure 3j–l).

Elemental mappings, energy-dispersive X-ray (EDX) analyses, inductively coupled plasma optical emission spectroscopy (ICP–OES), and thermogravimetric analysis (TGA) measurements were performed to further quantify the loading amount of Pt. Elemental mappings displayed that Pt, Fe, Ni, and O elements were distributed homogeneously (Figures S21–S23). EDX revealed that the Pt mass contents of Pt@MOF-BCP/BDC/NDC were 24.6%, 18.4%, and 14.5%, respectively (Figures S24–S26). ICP–OES analysis showed that the actual amounts of Pt clusters were 22.1, 16.0, and 8.2 wt % (Table S1). TGA measurements of Pt@MOF-BCP/BDC/NDC exhibited mass losses of 23.3%, 18.0%, and 13.2%, respectively (Figures S27–S29). Pt@MOF-BCP exhibited the highest dispersion density and greatest Pt loading within the pores, attributable to the minimal segment space of the MOF-BCP precursor. The Pt loading increased linearly as the pore sizes of MOF precursors decreased. All results demonstrated that the loading amount and particle size of Pt clusters are clearly regulated by the PSP process (Figure 3m).

The atomic structure of the Pt cluster was further characterized by aberration-corrected HAADF-STEM (AC HAADF-STEM). As shown in Figure 3n, bright spots were observed at the edge of Pt@MOF-BCP, representing the existence of Pt clusters. AC HAADF-STEM images displayed that the size of domination Pt clusters in Pt@MOF-BCP was

about 1.2 nm. The lattice spacings of 0.252 nm can be attributed to the (111) planes of Pt clusters (Figures S30 and S30). The lattice fringe of Pt clusters within Pt@MOF-BCP was slightly larger than the interplanar distance of intrinsic Pt (111) (0.226 nm),²⁹ attributed to the stretched coordination environment. The fast Fourier transform (FFT) image was assigned to metallic Pt (111), further confirming the formation of Pt clusters in partitioned pore space.

N₂ adsorption/desorption isotherms at 77 K showed Pt@MOFs with microporous features, illustrating their integral crystalline structures (Figures S31–S33). Compared with their pristine MOF precursors, the dramatically decreased specific surface areas and pore volumes of Pt@MOFs indicate that Pt incorporation within the pores induces partial pore occlusion, thereby confirming the strong spatial confinement effect and structural fastness of Pt within MOFs (Table S2).

The electronic structures of Pt@MOFs were verified by X-ray photoelectron spectroscopy (XPS) and X-ray absorption spectroscopy. The Pt 4f spectrum of Pt@MOF-BCP was located between K₂PtCl₄ and Pt foil, confirming the formation of metallic Pt and charge consumption on the Pt clusters side (Figure 4a).³⁰ With the pore space enlarged, the energy of Pt shifted to a higher energy, suggesting that there is more dispersity of Pt clusters. The discrete coordination increased the surface energy and shifted to higher electron binding energies.³¹ In the XPS spectra of N 1s, the peak at 399.2 eV was attributed to the pyridine N atoms in MOF-BCP.³² After introduction of Pt clusters, a new peak at 400.2 eV was observed in Pt@MOF-BCP, which can be assigned to the Pt–N bond.³³ The new peak further confirmed the newly generated coordination modes between Pt and the precursor MOFs (Figure 4b). Besides, the peak of pyridine N in Pt@MOF-BCP showed a negative shift, proving the electrons transfer from Pt clusters to the TPP linker, which was in accord with the above DFT calculations.

The normalized Pt L₃-edge X-ray absorption near edge structure (XANES) spectra of Pt@MOFs displayed that all of the white line intensity located between Pt foil and PtO₂, verifying the average valence state of Pt was between Pt⁰ and Pt⁴⁺ (Figure 4c). Among them, the white line intensity of Pt@MOF-BCP was much closer to Pt foil, suggesting the most metallic Pt.³⁴ According to the correlation of absorption edge, the average valence states of Pt were +0.05, +0.10, and +1.10 in Pt@MOF-BCP/BDC/NDC catalysts, respectively (Figure S34). The monitor trends mainly originated from a greater degree of electron overlap in Pt@MOF-BCP.³⁵ Pt L-edge EXAFS spectra of three Pt@MOFs exhibited major peaks at a distance of 2.1, 1.8, and 1.8 Å, respectively (Figure 4d). Particularly, the major peak of Pt@MOF-BCP showed the greatest degree of shift, which was attributed to the multiple coordination in the smallest pore space and the strongly interaction between Pt clusters and MOF-BCP.

Combined with structural design and theoretical calculation, the EXAFS fitting results indicated that the main peaks of Pt@MOFs were assigned to Pt–N and Pt-metal coordination (Figures 4e and S35–S36). On account of the most electronegativity difference, Pt atoms contact N atoms from parallel TPP linkers to form Pt–N bonds, which was reasonable. Furthermore, EXAFS fitting of Pt@MOF-BCP/BDC/NDC confirmed the lengths of Pt–N were 1.96, 1.98, and 2.03 Å, respectively, stating the different atomic coordination configuration of Pt–N in different pore spaces (Table S3). Benefiting from the smallest pore size and the

strong conjugated system between Pt clusters and MOF support, the Pt–N bonds in Pt@MOF-BCP possessed the shortest bond length, which was also consistent with the DFT calculation.

Remarkably, except for the formation of major coordination Pt–N bonds, Pt-metal coordination was also formed in Pt@MOFs. Pt-metal coordination configuration was further surveyed using quantitative least-squares EXAFS curve-fitting.³⁶ EXAFS fitting of Pt@MOF-BCP exhibited the predominant Pt–Fe/Ni (2.52 Å) and Pt–Pt (2.66 Å) configurations. That is, due to the interconnected partitioned pore space of MOF-BCP, Pt can interact with the metal nodes of the MOF-BCP support to form Pt–Fe/Ni bonds. Besides, the inherent infinity frameworks of MOF-BCP provided the chance to cross-link the adjacent Pt atoms to form the infinite 1D Pt array. Above all, as shown in Figure 4f, each Pt atom can interact with aromatic TPP partitioners, Fe/Ni nodes of intrinsic MOF, and adjacent Pt atoms simultaneously to give a triple coordination configuration.

To more precisely clarify the coordination conditions of Pt, WT analysis was carried out. Especially, the intensity maximum at 6.6 Å^{−1} for Pt@MOF-BCP located between PtO₂ and Pt foil, which is different from that for the Pt–O (5.6 Å^{−1}) as well as Pt–Pt coordination mode (9.4 Å^{−1}), further confirms the intercoordination conditions for Pt in MOF-BCP. With the pore spaces of precursor MOFs increased, triple coordination configurations trend to clearness. This may be caused by the more discrete Pt clusters in the pore and the more distinguishable average coordination (Figure 4g). All these results successfully resolve the precise coordination configuration and atomic-level structural details of the sandwich-structured Pt array encapsulated within Pt@MOFs.

Fe 2p and Ni 2p XPS spectra of three Pt@MOFs all showed positive shifts compared with MOF supports, implying the electron transfer from the Fe/Ni to TPP linker (Figures S37 and S38). Meanwhile, the higher oxidation states of both Fe and Ni were observed within the smaller pores, which were attributed to enhanced electronic interactions between the contracted MOF pore architecture and the encapsulated Pt clusters (Figure S39). The energy absorption edge of the Pt@MOF-BCP catalyst was located at the position near Fe₂O₃, verifying the existence of Fe³⁺ (Figure S40). EXAFS spectra of Fe validated that the typical Fe–O(N) scattering configuration was located at ~1.4 Å with a slight shift, further implying the interaction between Pt and the intrinsic MOF-BCP (Figure S41). Ni K-edge XANES spectra of Pt@MOF-BCP showed features similar to those of NiO, implying a +2 oxidation state similar to that of NiO (Figure S42). Similar to Fe, the main peak at 1.5 Å corresponding to Ni–O coordination showed a slight shift (Figure S43). EXAFS fitting results of Fe/Ni in Pt@MOF-BCP suggested that except the normal coordination of metal (Fe/Ni)–O in nodes of intrinsic MOF, the coordination between Pt and Fe/Ni was formed undoubtedly (Figures S44–S45 and Tables S4–S5).

HER and OER performances of Pt@MOF catalysts were investigated in 0.5 M H₂SO₄ and 1 M KOH electrolyte to reveal the facilitation of catalytic reactions. Cyclic voltammetry (CV) curves were measured in a N₂-saturated 0.5 M H₂SO₄ solution. Compared with the precursor MOF-BCP, an obvious hydroxyl adsorption/desorption peak can be observed on Pt@MOF-BCP, which can be attributed to the stronger H* intermediates' adsorption energy on Pt@MOF-BCP (Figure 5a).³⁷ Deriving from the sandwich Pt arrays within MOFs, the

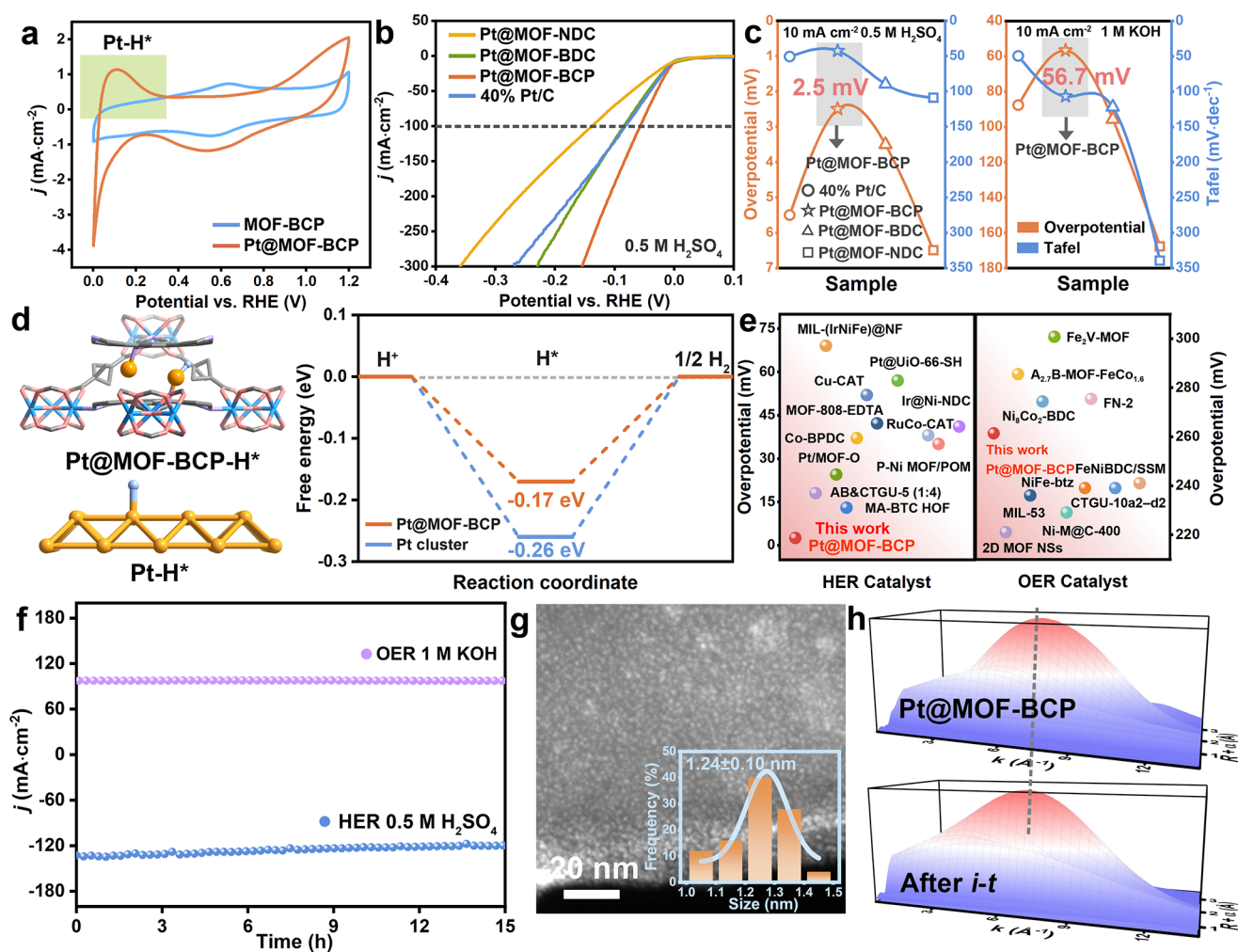


Figure 5. (a) CV curves of Pt@MOF-BCP and MOF-BCP in a N₂-saturated atmosphere. (b) LSV curves of the series Pt@MOFs and 40% Pt/C for HER in 0.5 M H₂SO₄. (c) Comparison of overpotential and Tafel slopes on Pt@MOFs. (d) Free energy of adsorbed H* intermediates on the Pt@MOF-BCP and Pt cluster. (e) Comparison of other similar catalysts. (f) *i*-*t* test for HER and OER. (g) HAADF-STEM image after HER. (h) Wavelet transforms for the *k*₃-weighted EXAFS of Pt@MOF-BCP after the *i*-*t* test for HER.

local charge distribution in Pt@MOFs was enhanced, the electron supply capacity was promoted, and then the energies of H* intermediate adsorption and H₂ evolution were determined.

Among them, Pt@MOF-BCP exhibited excellent HER performance in 0.5 M H₂SO₄. Linear sweep voltammetry (LSV) curves demonstrated that the overpotential of Pt@MOF-BCP at 10 mA·cm⁻² was 2.5 mV. Moreover, to reach the high current density of 100 mA·cm⁻², the overpotential of only 57.5 mV is required for Pt@MOF-BCP, which is much lower than those required for Pt@MOF-BDC (85.5 mV), Pt@MOF-NDC (138.5 mV), Pt@MIL-88 (274.5 mV), and even Pt/C (82.5 mV) (Figures 5b and S46–S47). Notably, Pt@MOF-BCP still exhibited the lowest onset potential, the highest BET-normalized current density, and a pronounced enhancement in turnover frequency (TOF) value and mass activity, highlighting its superior intrinsic HER activity (Figure S48 and Table S6).

The Tafel slope of Pt@MOF-BCP (42.5 mV·dec⁻¹) was significantly lower than those of Pt@MOF-BDC and Pt@MOF-NDC, consistent with a Volmer–Heyrovsky rate-determining step, confirming the more favorable reaction kinetics for H* adsorption on Pt@MOF-BCP (Figure

S49).^{38,39} In addition, Pt@MOF-BCP showed the smallest charge transfer resistance (*R*_{ct}) of 18.04 Ω, which is 1.3 and 2.2-fold diminution for Pt@MOF-BDC (22.79 Ω) and Pt@MOF-NDC (39.64 Ω) (Figure S50). Pt@MOF-BCP displayed the largest electric double-layer capacitance *C*_{dl} value of 76.6 mF·cm⁻², which is 1.2 and 7.0-fold enhancement over Pt@MOF-BDC (62.1 mF·cm⁻²) and Pt@MOF-NDC (11.0 mF·cm⁻²), respectively (Figure S51). Clearly, the Pt@MOF-BCP catalyst exhibits most efficient charge-transfer kinetics and largest electrochemically active area. The consistent trends in electrochemical performance, including *η*₁₀, Tafel slope, *C*_{dl}, and *R*_{ct}, suggested that the optimized Pt@MOF-BCP can facilitate favorable reaction kinetics for H* adsorption and conversion (Figure S52).

HER performance was also investigated in the alkaline electrolyte. Pt@MOF-BCP displayed the most excellent HER performance in acidic and alkaline electrolytes simultaneously (Figures 5c and S53–S56). Pt clusters within packaged MOFs were considered as active species for HER. Variations in the confined pore spaces of Pt@MOFs resulted in distinct spatial distributions and host–guest interaction forces between the Pt cluster and MOF frameworks via $\pi \cdots \text{Pt} \cdots \pi$ electronic conjugated coordination. Pt@MOF-BCP, characterized by

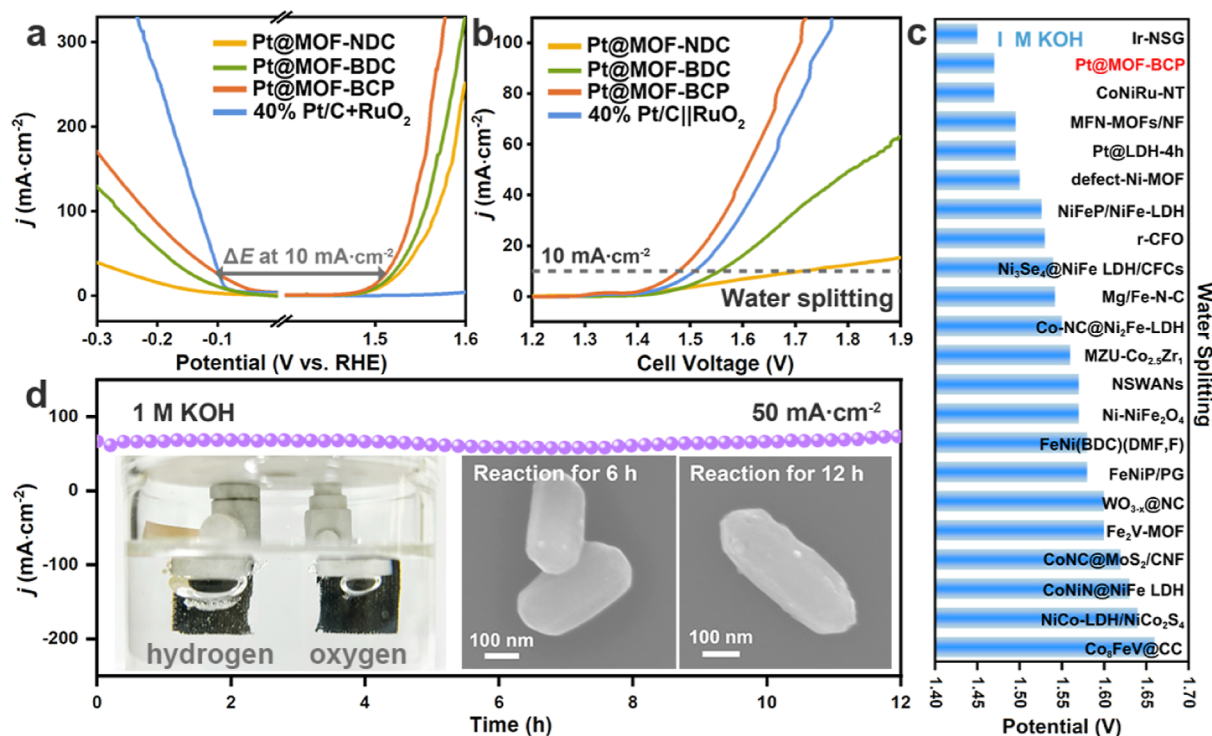


Figure 6. (a) Value of ΔE for Pt@MOFs and Pt/C + RuO₂ in 1 M KOH for HER and OER. (b) LSV curves for overall water splitting in 1 M KOH. (c) Comparison of overall water splitting activity. (d) *i*-*t* test at 50 mA·cm⁻² for 12 h.

minimal Pt–N(TPP) distances (1.94 Å), exhibited the strongest electronic interaction between Pt and the framework (Figure 4a). Furthermore, the inevitable interplay between Pt and pristine Fe/Ni metal clusters within the MOF-BCP frameworks also modulated the electronic structure, as evidenced by XPS results (Figure S39). The synergistic electronic optimization of Pt enhances electronic coupling, accelerates interfacial charge transfer, and stabilizes the H* intermediates during HER catalysis.

To elucidate the intrinsic relationship between π ···Pt··· π coordination and enhanced electrocatalytic activity, DFT calculations were further performed on Pt@MOF-BCP and Pt clusters in the acidic electrolyte (Figures S57–S59). DFT calculations revealed that ΔG_{H^*} at Pt@MOF-BCP was −0.17 eV, which was closer to the ideal thermoneutral value of $\Delta G_{H^*} = 0$ eV compared to the standard Pt cluster of ΔG_{H^*} of −0.26 eV,⁴⁰ indicating the more favorable adsorption/desorption kinetics for a key intermediate of H* (Figure Sd).⁴¹ Therefore, due to the high density of Pt sites and optimized electronic structure within Pt@MOF-BCP, the greatest extent modulation of intermediate H* binding energy through the π ···Pt··· π electronic conjugate system facilitates the H₂ evolution.

Furthermore, the different loading amounts of Pt were further explored. Pt@MOF-BCP catalysts with different input amounts of Pt were prepared. The excessive concentration would result in the agglomeration of Pt clusters, thereby compromising their catalytic efficacy. The insufficient amount of Pt leads to the inadequate modulation of electronic structure and ultimately impairs the desired catalytic performance (Figures S60–S70).

OER performances of Pt@MOFs were tested in 1 M KOH. It can be seen that to reach a high current density of 100 mA·cm⁻², the potential of Pt@MOF-BCP was 1.54 V (Figures S71 and S72), which was lower than those of Pt@MOF-BDC (1.55

V), Pt@MOF-NDC (1.56 V), RuO₂ (1.8 V), Pt/C (1.75 V), and Pt@MIL-88 (1.67 V) catalysts. Compared with commercial Pt- and Ru- based catalysts, Pt@MOF-BCP exhibited superior intrinsic activity, achieving a TOF value of 0.043 s⁻¹ and mass activity of 42.1 A·g⁻¹ at $\eta = 300$ mV during OER (Table S7). Bode plots of Pt@MOF-BCP exhibited the greatest degree of impedance, further indicating the fast charge-transfer kinetics (Figure S73).⁴² Pt@MOF-BCP exhibited the a smallest Tafel slope, smallest *R*_{ct} value, and the largest *C*_{dl} value during the OER reaction on account of the outperforming reaction kinetics (Figures S74–S76). The superior OER performance of Pt@MOF-BCP is attributed to the stabilization of high-valent metal centers (Fe³⁺/Ni³⁺), which is facilitated by the stronger Pt-support interactions compared to Pt@MOF-BDC and Pt@MOF-NDC. This enhanced charge polarization of metal centers promotes the formation of active intermediates, directly contributing to catalytic enhancement.⁴³

During the OER process, DFT calculations showed that the energy barrier of the O* formation step was 0.59 eV for Pt@MOF-BCP, which was smaller than that of the pristine Pt cluster (0.71 eV), indicating the improvement of electronic structure via Pt-support interactions (Figures S77–S80). Thus, the newly generated π ···Pt··· π electronic conjugate system enhances the intrinsic reactive kinetics for water oxidation.^{44,45} Hence, Pt@MOF-BCP exhibited best performance for HER and OER in acidic and alkaline electrolytes, which was superior to nearly all similar catalysts (Figure 5e and Table S8).^{46,47}

Catalytic durability plays a crucial role in practical applications. Profiting from the unique sandwich Pt array within pore-space-partitioned MOFs, Pt@MOF-BCP also exhibited superior stability for HER and OER in acidic and alkaline electrolytes (Figure 5f). The XRD revealed that the structure of the MOF was maintained after HER (Figure S81).

Pt clusters were still confined in the pore spaces with slight agglomeration, and the size was about 1.24 ± 0.1 nm (Figures Sg and S82–S83). Postreaction EDX, TGA, and ICP-OES analyses revealed the negligible percentage composition of Pt loading after HER reaction, indicating the structural stability of a sandwich Pt array within the MOF (Figure S84). Furthermore, ICP-OES analysis of the post-HER electrolyte detected no leaching of Pt species, conclusively demonstrating the structural robustness of Pt within MOFs and further substantiating exceptional electrochemical durability. Furthermore, XPS, normalized XANES spectra, and WT analysis indicated the slight shift after HER reaction (Figures Sh and S85–S86). EXAFS fitting results showed that the bond length of Pt–N was 1.82 Å, slightly shortened after HER, demonstrating the slight variation of local electronic structures (Figure S87 and Table S9), which may be caused by the mild agglomeration of Pt clusters. Overall, the Pt@MOF-BCP catalyst exhibited excellent activity and robust stability, making it possible to be utilized in practical overall water splitting application (Figures S88–S98).

Inspired by the exceptional HER and OER performance, two-electrode electrolyzer tests were carried out in 1 M KOH with Pt@MOFs as the cathode and anode. The ΔE value of Pt@MOF-BCP was about 1.55 V at $10 \text{ mA}\cdot\text{cm}^{-2}$, which was lower than those of Pt@MOF-BDC (1.60 V), Pt@MOF-NDC (1.67 V), and Pt/C + RuO₂ (1.72 V), indicating the excellent performance in two-electrode tests (Figure 6a). The cell voltage of Pt@MOF-BCP||Pt@MOF-BCP (1.47 V) reached the current density of $10 \text{ mA}\cdot\text{cm}^{-2}$, which was significantly lower than those of Pt@MOF-BDC (1.55 V), Pt@MOF-NDC (1.71 V), and Pt/C||RuO₂ (1.50 V), indicating outstanding electrocatalytic activity of Pt@MOF-BCP for overall water splitting in practice (Figure 6b). The Tafel slope of Pt@MOF-BCP||Pt@MOF-BCP proved to possess superior reaction kinetics for water electrolysis in 1 M KOH (Figure S99). Encouragingly, the bifunctional Pt@MOF-BCP catalyst exhibited extraordinary performance toward overall water splitting in alkaline media, which was superior to most MOF-based catalysts (Figure 6c). Long-term stability of the bifunctional Pt@MOF-BCP catalyst was further examined, which showed superior electrocatalytic stability in two-electrode in alkaline media (Figure 6d). Large amounts of H₂ and O₂ gas products can be generated on the cathode and anode devices simultaneously, indicating the competent and persistent catalyst for overall water splitting. Faradaic efficiency of overall water splitting reaction over the Pt@MOF-BCP catalyst was calculated to be around 97% under the current density of $100 \text{ mA}\cdot\text{cm}^{-2}$ for 120 min (Figure S100). SEM, TEM, and HAADF-STEM images for the catalyst after overall water splitting further revealed the ultrarobust structure of Pt@MOF-BCP under high current density (Figures 6d inset and S101–S102).

CONCLUSION

In this study, a series of precisely engineered sandwich-structured platinum arrays were successfully constructed within pore-space-partitioned MOFs. Platinum was encapsulated within two parallel aromatic partitioners, forming a well-defined sandwich architecture. The size of Pt clusters was systematically controlled by modulating the geometry of ditopic linkers, while a stable $\pi\cdots\text{Pt}\cdots\pi$ conjugated electronic system was established through interactions between π -electrons from the partitioners and d-orbitals of Pt. Notably,

Pt clusters located in adjacent pores lined up to form continuous 1D Pt arrays extending along the partitioned MOF channel. The spatial confinement stabilized the sandwich Pt arrays and modified their electronic structure. The optimized catalyst, Pt@MOF-BCP (derived from the smallest pore size), demonstrated exceptional electrocatalytic performance with an overpotential of 2.5 mV for HER in the acidic electrolyte and 265 mV for OER in the alkaline electrolyte at $10 \text{ mA}\cdot\text{cm}^{-2}$ as well as top-level overall water splitting activity and stability superior to benchmark catalysts. This work provides a paradigm for the atomic-level design of noble-metal secondary structures within MOFs, offering new avenues for advanced energy conversion technologies.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c04935>.

Experimental details; DFT calculation details; morphology and structure characterizations; and electrochemical data (PDF)

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Notes

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